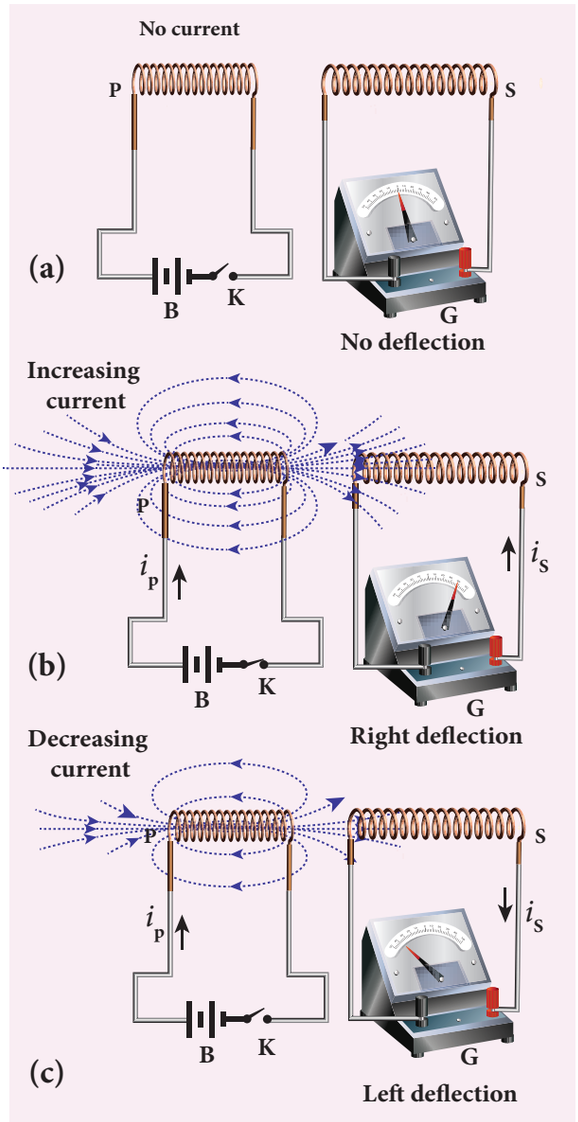


Figure 4.5 Explanation of Faraday's second experiment

magnetic field around the primary coil. Therefore, the magnetic flux linked with the secondary coil increases. This increasing flux linked induces a transient electric current in the secondary coil (Figure 4.5(b)). When the electric current in the primary coil reaches a steady value, the magnetic flux linked with the secondary coil does not change and the electric current in the secondary coil will disappear.

Similarly, when the primary circuit is broken, the decreasing primary current induces an electric current in the secondary coil, but in the opposite direction (Figure 4.5(c)). So there is deflection in the galvanometer whenever there is a change in the primary current.

The conclusions of Faraday's experiments are stated as two laws.

First law

Whenever magnetic flux linked with a closed circuit changes, an emf is induced in the circuit which lasts in the circuit as long as the magnetic flux is changing.

Second law

The magnitude of induced emf in a closed circuit is equal to the time rate of change of magnetic flux linked with the circuit.

If the magnetic flux linked with each turn of the coil changes by $d\Phi_B$ in a time dt, then the induced emf in each turn is given by

$$\varepsilon = \frac{d\Phi_B}{dt}$$

If a coil consisting of N turns is tightly wound such that each turn covers the same area, then the flux through each turn will be the same. Then total emf induced in the coil is given by

$$\varepsilon = N\frac{d(\Phi_B)}{dt} = \frac{d(N\Phi_B)}{dt} \qquad (4.2)$$

Here $N\Phi_B$ is called flux linkage, defined as the product of number of turns N of the coil and the magnetic flux linking each turn of the coil Φ_B.

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